

RYAN E. BARSZCZ

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SUMMARY

Software Engineer with experience designing full-stack web applications, desktop software, and real-time data systems. Built an end-to-end vehicle telemetry platform integrating OBD-II hardware, Python, FastAPI, PostgreSQL, and Next.js. Passionate about building scalable software that bridges modern web technologies with real-world systems.

SKILLS

- **Languages:** TypeScript, JavaScript, Python, Java, C++
- **Frontend:** React, Next.js, Tailwind CSS, GSAP, React Native, Flutter
- **Backend & Data:** FastAPI, Node.js, Express, SQLAlchemy, Prisma, PostgreSQL, MongoDB
- **Cloud & DevOps:** Docker, Docker Compose, AWS, GitHub Actions, Vercel, Neon, Firebase, PyInstaller
- **Testing & Tools:** Git, Postman, CI/CD, pytest, FastAPI TestClient, CMS, Figma, Photoshop
- **Certifications:** Google Generative AI Fundamentals (2023)
- **Professional Skills:** Full Stack Development, Software Engineering, Software Testing

EDUCATION

University of Michigan—Ann Arbor
B.S., Computer Science

Jun 2025

PROJECT EXPERIENCE

DriveIQ - Vehicle Telemetry Platform <https://telemetry.ryanbarszcz.com/> **Jun 2026 - Jul 2026**

- Built an end-to-end OBD-II vehicle telemetry platform using Next.js, TypeScript, FastAPI, Python, SQLAlchemy, PostgreSQL, Docker, Docker Compose, Clerk, AWS S3, and Vercel to capture, visualize, and analyze live and historical vehicle telemetry data.
- Developed and packaged a Windows Python logger that connects to a USB OBD-II adapter, samples configurable metrics, records session CSVs, and uploads completed drives to an authenticated cloud API.
- Implemented synchronized telemetry charts, authenticated vehicle/session management, CSV-based analysis, and secure S3-backed file upload and download workflows
- Containerized the Next.js frontend, FastAPI backend, and PostgreSQL database with Docker Compose, creating a reproducible multi-service development environment with persistent storage and automated networking.
- Added 25 automated API and integration tests with pytest and FastAPI TestClient, including mocked S3 workflows, and built GitHub Actions CI to run tests and verify Docker builds on every push and pull request.

SetupsRus - Sim Racing Setup Sharing Platform <https://www.setupsrus.com/> **Jun 2026**

- Built a full-stack sim racing setup-sharing platform using Next.js, TypeScript, Node.js, Express, Prisma, and PostgreSQL, enabling users to upload, browse, and download vehicle setup files.
- Developed secure authentication and ownership-based permissions with Clerk, supporting user profiles, setup management, and voting functionality.
- Integrated AWS S3 object storage and presigned URL generation for secure file uploads and downloads, enabling scalable cloud-based asset management.
- Designed and implemented RESTful APIs and relational database schemas for games, tracks, vehicles, tags, and user-generated content, supporting advanced filtering, search, and content discovery.

Alert Arrival - Best Automotive Hackathon Winner

Mar 2024

GrizzHacks 6

- Prototyped a driver fatigue detection system using OpenCV eye-aspect-ratio analysis to assess drowsiness in real time. Built a Flutter UI to visualize telemetry and streamed live alerts through Firebase. Awarded Best Automotive Hack among 40+ teams for demonstrating a high-impact safety-focused mobility concept.

Liberty Athletic Club Website

Sep 2025 - Oct 2025

Liberty Athletic Club

- Designed and developed a modern website for Liberty Athletic Club using Next.js, React, Tailwind CSS, and GSAP, creating a polished, responsive experience for an organization serving over 7,000 members.

EXPERIENCE

Liberty Athletic Club

Feb 2023 - Present

Aquatics Director Assistant / Club Manager

- Oversee daily operations of the high-traffic athletic facility, coordinating staff with scheduling software and resolving issues in real time, which improves member wait times and keeps the facility running smoothly.
- Lead on-shift safety coordination, incident response, and staff communication.
- Recognized as Employee of the Month for leadership and reliability 5+ times.